



Draft genome sequence of a carbapenem-resistant clinical *Acinetobacter baumannii* revealing co-existence of four classes of β -lactamases

M. Ishrat Jahan, Md. Mizanur Rahaman, M. Anwar Hossain¹, Munawar Sultana*

Department of Microbiology, University of Dhaka, Dhaka, 1000, Bangladesh

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ABSTRACT

Objectives: Nosocomial carbapenem-resistant *Acinetobacter baumannii* is a challenge in the treatment of intensive care unit (ICU) patients. The presence of antimicrobial resistance genes (ARGs) and mobile genetic elements can further complicate effects to combat antibiotic resistance in this post-antibiotic era. The aim of this study was to analyse the molecular basis of carbapenem resistance in *A. baumannii* strain DMC-32a.

Methods: Strain DMC-32a, isolated from a wound swab of an ICU patient, was screened phenotypically and genotypically for carbapenem resistance. The isolate was subjected whole-genome sequencing (WGS) to investigate the resistance mechanisms.

Results: Strain DMC-32a was resistant to all tested antibiotics belonging to seven classes, except for the polymyxins. MICs determined against imipenem and meropenem were 512 mg/L and >512 mg/L, respectively. Gene-specific PCR confirmed the presence of *bla*_{NDM-1} and *int1*. WGS confirmed the isolate as sequence type ST1053, the presence of four classes of β -lactamases [A (*bla*_{PER-7}, *bla*_{TEM-116}), B (*bla*_{NDM-1}), C (*bla*_{ADC-26}) and D (*bla*_{OXA-23}, *bla*_{OXA-51})], and four genes encoding aminoglycoside-modifying enzymes [*aph*(6)-Ia, *aph*(3'')-Ib, *aph*(3'')-VI and *ant*(3'')-IIc] conferring resistance to streptomycin, amikacin, kanamycin and spectinomycin. The analysis also confirmed the presence of a class 1 integron gene cassette harbouring ARGs to rifamycin (*arr-2*) and phenicols (*cmlA5*). No plasmid replicon was found from the sequence data.

Conclusion: The co-existence of four β -lactamase classes in *A. baumannii* DMC-32a has not been reported previously from Bangladesh and not in this ST elsewhere. The emergence of such a nosocomial pathogen creates the need for effective therapeutics for critically-ill patients and for controlling hospital-acquired infections.

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1. Introduction

Carbapenem-resistant *Acinetobacter baumannii*, one of the members of the nosocomial pathogen group 'ESKAPE' (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa* and *Enterobacter* spp.), has been categorised as a priority 1 critical pathogen by the World Health Organization (WHO) in 2017. Among the carbapenem resis-

tance mechanisms, production of β -lactamases is the most common. According to the Ambler classification, there are four classes of β -lactamases (A, B, C and D). Among these, classes A, B and D are called carbapenemases owing to their ability to hydrolyse carbapenems. The presence of AmpC β -lactamase and OXA-51 makes *A. baumannii* intrinsically resistant to some β -lactams. The involvement of class 1 integrons along with antimicrobial resistance genes (ARGs) creates a global concern for the dissemination of antimicrobial resistance. To understand the resistance profile and the involvement of the mobilome in disseminating resistance, *A. baumannii* strain DMC-32a was analysed by whole-genome sequencing (WGS).

* Corresponding author. Tel.: +880 2 966 192 073 #7731.

E-mail address: munawar@du.ac.bd (M. Sultana).

¹ Present address: Vice-Chancellor, Jessore University of Science and Technology, Jessore 7408, Bangladesh.

Table 1Antimicrobial resistance genes (ARGs) found in carbapenem-resistant *Acinetobacter baumannii* strain DMC-32

ARG family	ARO term	Target antibiotic(s)	Location in WGS
Bleomycin resistance protein	BRP (MBL)	Glycopeptide	NODE_76
TEM β -lactamase	TEM-116	Monobactam, cephalosporin, penam, penem	NODE_107
PER β -lactamase	PER-7	Monobactam, carbapenem, cephalosporin, penam, penem	NODE_72
NDM β -lactamase	NDM-1	Carbapenem, cephalosporin, cephamycin, penam	NODE_76
AmpC-type β -lactamase	ADC-26	Cephalosporin, penam	NODE_1
OXA β -lactamase	OXA-104	Cephalosporin, penam	NODE_2
	OXA-23	Cephalosporin, penam	NODE_100
Resistance–nodulation–cell division (RND) antibiotic efflux pump	AcrB	Tetracycline, phenicol, rifamycin, glycylcycline, penam, cephalosporin, triclosan, fluoroquinolone	NODE_13
	AdeK	Macrolide, fluoroquinolone, lincosamide, carbapenem, cephalosporin, tetracycline, rifamycin, diaminopyrimidine, phenicol, penem	NODE_6
	AdeJ	Macrolide, fluoroquinolone, lincosamide, carbapenem, cephalosporin, tetracycline, rifamycin, diaminopyrimidine, phenicol, penem	NODE_6
	AdeI	Macrolide, fluoroquinolone, lincosamide, carbapenem, cephalosporin, tetracycline, rifamycin, diaminopyrimidine, phenicol, penem	NODE_6
	AdeL	Fluoroquinolone, tetracycline	NODE_1
	AdeF	Fluoroquinolone, tetracycline	NODE_1
	AdeN	Macrolide, fluoroquinolone, lincosamide, carbapenem, cephalosporin, tetracycline, rifamycin, diaminopyrimidine, phenicol, penem	NODE_3
	AdeT	Aminoglycoside	NODE_3
	AdeS	Glycylcycline, tetracycline	NODE_3
	AdeR	Glycylcycline, tetracycline	NODE_3
	AdeA	Glycylcycline, tetracycline	NODE_3
	AdeB	Glycylcycline, tetracycline	NODE_3
	CzcA, CzcB, CzcC	Cobalt, zinc, cadmium	NODE_18
Major facilitator superfamily (MFS) antibiotic efflux pump	<i>Acinetobacter baumannii</i> AbaQ	Fluoroquinolone	NODE_3
	<i>Acinetobacter baumannii</i> AbaF	Fosfomycin	NODE_12
	EmrB	Fluoroquinolone	NODE_24
	EmrB	Fluoroquinolone	NODE_24
	CmlA5	Phenicol	NODE_72
	<i>Acinetobacter baumannii</i> AmvA	Macrolide, acridine dye	NODE_14
	MdfA	Chloramphenicol, erythromycin, certain aminoglycosides, rhodamine, fluoroquinolone, daunomycin, benzalkonium, rifampicin, tetracycline, puromycin	NODE_15
	Cmr	Tetracycline, benzalkonium chloride, rhodamine	NODE_15
ATP-binding cassette (ABC) antibiotic efflux pump	MacA	Macrolide	NODE_10
Small multidrug resistance (SMR) antibiotic efflux pump	AbeS	Macrolide, aminocoumarin	NODE_1
Multidrug and toxic compound extrusion (MATE) family efflux pump	MdtK/NorM	Fluoroquinolone	NODE_26
FosA glutathione transferase	FosA	Fosfomycin	NODE_65
Rifampin ADP-ribosyltransferase (Arr)	Arr-2	Rifamycin	NODE_72
Macrolide phosphotransferase (MPH)	MphE	Macrolide	NODE_79
ABC-F ATP-binding cassette ribosomal protection protein	MsrE	Macrolide, lincosamide, streptogramin, tetracycline, oxazolidinone, phenicol, pleuromutilin	NODE_79
16S rRNA methyltransferase (G1405)	ArmA	Aminoglycoside	NODE_79
ANT(3'')	ANT(3'')-IIc	Aminoglycoside	NODE_28
APH(6)	APH(6)-Id	Aminoglycoside	NODE_70
APH(3'')	APH(3'')-Ib	Aminoglycoside	NODE_70
APH(3'')	APH(3'')-VI	Aminoglycoside	NODE_215

ARO, Antibiotic Resistance Ontology; WGS, whole-genome sequencing.

2. Materials and methods

Acinetobacter baumannii DMC-32a, previously isolated from a wound swab of an intensive care unit patient [1], was screened phenotypically for resistance to 15 antibiotics from seven classes according to Clinical and Laboratory Standards Institute (CLSI) M100S guidelines using the Kirby–Bauer disk diffusion method: ce-

fepime (30 μ g); cefotaxime (30 μ g); ceftazidime (30 μ g); ceftriaxone (30 μ g); doxycycline (30 μ g); tetracycline (30 μ g); gentamicin (10 μ g); ciprofloxacin (5 μ g); levofloxacin (5 μ g); imipenem (10 μ g); meropenem (10 μ g); doripenem (10 μ g); trimethoprim/sulfamethoxazole (25 μ g); polymyxin B (10 μ g); and colistin (10 μ g) (Oxoid Ltd., Basingstoke, UK). Minimum inhibitory concentrations (MICs) were determined for carbapenems (imipenem

and meropenem) by the broth microdilution method. Gene-specific PCR was used to determine the presence of carbapenemase genes (*bla_{KPC}*, *bla_{VIM}*, *bla_{NDM}*, *bla_{IMP}* and *bla_{OXA}*) and *int11*, a marker of class 1 integrons [2,3]. Total DNA was extracted using a QIAamp® UCP Pathogen Mini Kit (QIAGEN, Hilden, Germany) according to the manufacturer's instructions. Whole genomic DNA was subjected to WGS using a high-throughput ion semiconductor Ion-Torrent™ at Thermo Fisher Institute, India. FastQC v.0.111 (Andrew, 2010) was used to analyse the quality of reads. Raw data were assembled using SPAdes v.3.13.0 and were annotated using RAST (<http://rast.nmpdr.org/>). To understand the genomic epidemiology, we used BacWGSTdb 2.0 (<http://bacddb.cn/BacWGSTdb>) that uses both core genome multilocus sequence typing (cgMLST) and a single nucleotide polymorphism (SNP)-based scheme for genomic comparison [4]. The draft genome sequence was submitted to NCBI under accession no. **SRJ00000000.2**.

3. Results and discussion

Strain DMC-32a was resistant to all of the tested antibiotics except colistin and polymyxin B. The MICs against imipenem and meropenem were 512 mg/L and >512 mg/L, respectively. Sequencing data confirmed the presence of *bla_{NDM-1}* and *int11* genes.

The draft genome was 4 386 240 bp (281 contigs; *N*₅₀ 127 309 bp) with 4304 coding sequences (CDS) and a GC content of 39.3%. From the BacWGSTdb 2.0 database, the sequence type determined was ST1053, and genomic comparison revealed no closely related strain indicating the probability of it being a sporadic strain. Both CARD: RGI (<https://card.mcmaster.ca/analyze/rgi>) and ResFinder 3.0 (<http://www.genomicepidemiology.org/>) confirmed the presence of an array of ARGs (Table 1). All four classes of β -lactamases were found to be present. Class A [extended-spectrum β -lactamase (ESBL)]: PER-7 (NODE_72) and TEM-116 (NODE_107) possess higher-level hydrolytic activities against cephalosporins and aztreonam. Class B: NDM-1 (NODE_76), a novel carbapenemase, can hydrolyse all β -lactams except aztreonam. Class C: ADC-26 (NODE_1), when chromosomally inherited with *ISAb1* in the upstream region, increases resistance to ceftazidime, cefotaxime and aztreonam if overproduced. Class D: OXA-23 (NODE_100) is considered a major cause of carbapenem resistance owing to its rapid dissemination and can be both plasmid-borne or chromosomally inherited in *A. baumannii* [5]. The co-existence of four β -lactamase classes in *A. baumannii* triggers an alarming situation in terms of nosocomial infections. To our knowledge, such co-existence is not reported in this sequence type from anywhere and so far there is none in any *A. baumannii* from Bangladesh. A similar kind of co-existence of four β -lactamase classes in carbapenem-resistant *P. aeruginosa* was reported in another study [6].

In addition to carbapenemases, strain DMC-32a contained several ARGs (Table 1). There were three genes encoding aminoglycoside phosphotransferases [*aph(6)-Id*, *aph(3'')-Ib* and *aph(3')-VI*] and one encoding an aminoglycoside nucleotidyltransferase [*ant(3'')-IIc*]. Both APH(6)-Id and APH(3'')-Ib confer resistance to streptomycin, APH(3')-VI confers resistance to amikacin and kanamycin, and ANT(3'')-IIc confers resistance to streptomycin and spectinomycin in *A. baumannii* [5]. Strain DMC-32a also harboured ARGs against bleomycin (an essential medicine according to the WHO), fosfomycin and rifampicin.

The sequence (NODE_72) confirmed a partially completed gene cassette structure, *int11-arr-2-cmlA5-qacE Δ 1-sul2* (3669 bp) (Supplementary Fig. S1) in the class 1 integron. A similar kind of resis-

tance gene cassette structure has been reported previously (GenBank **CP021344**). Although the isolate harboured plasmids when extracted using QIAprep® Spin Miniprep Kit (QIAGEN), no plasmid sequence was retrieved from the whole genome by PlasmidFinder (<https://cge.cbs.dtu.dk/services/PlasmidFinder/>). Vertical or horizontal transmission of this integron can provide resistance to aminoglycoside and phenicol antibiotics, quaternary ammonium compounds and sulfonamides.

4. Conclusion

Being susceptible only to colistin and polymyxin B, carbapenem-resistant *A. baumannii* strain DMC-32a highlights the antibiotic ineffectiveness nightmare in the post-antibiotic era in Bangladesh. Moreover, the presence of a resistance integron can contribute greatly to the transmissibility of such traits in clinical settings. Therefore, an effective solution to treat severely ill patients infected with such a superbug is warranted.

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Competing interests

None declared.

Ethical approval

This study was reviewed by the Ethical Committee of the Faculty of Biological Science, University of Dhaka, Bangladesh [Ref.64/Biol.Sc./2018-2019].

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jgar.2021.11.002](https://doi.org/10.1016/j.jgar.2021.11.002).

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